

Global Beats, Local Tongue

STUDYING CODE SWITCHING
IN K-POP HITS ON BILLBOARD CHARTS

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Outline

- Background and Motivation
- Code Switching vs Code Mixing
- Data collection
- Statistical Test
- Conclusion and Future Direction

BACKGROUND AND MOTIVATION



As 'Golden' Remains at No. 1, 'KPop Demon Hunters' Now Has Four Songs in the Hot 100's Top 10 — a Historic First for a Soundtrack

By Chris Willman



Billboard launches K-pop music charts for first time to track genre's local and global metrics



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KPop Demon Hunters Earns Three Golden Globe Nominations



Huntr/X don't miss. KPop [Demon Hunters](#), directed by [Maggie Kang](#) and [Chris Appelhans](#), continues its global domination that started with the film's launch on [Netflix](#) in June 2025, and hasn't stopped going up, up, up since. The animated musical epic has nabbed three nominations for the 83rd annual Golden Globes: Best Animated Motion Picture, Best Cinematic and Box Office Achievement, and Best Original Song from a Motion Picture for "Golden," with music by [Joong Gyu Kwak](#), [Yu Han Lee](#), [Hee Dong Nam](#) (Ido), [Jeong Hoon Seo](#) (24), [Park Hong Jun](#) (Teddy); and lyrics by [Ejae](#) and [Mark Sonnenblick](#).

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BTS EARNS FIRST NO. 1 ALBUM ON BILLBOARD 200 CHART WITH 'LOVE YOURSELF: TEAR'

Korean boy band BTS debuts at No. 1 on the Billboard 200 albums chart with its latest effort, "Love Yourself: Tear." It's the first No. 1 for the seven-member group, and the first K-pop album to lead...

By **Keith Caulfield** ▾

BLACKPINK'S MAJOR DEBUT: NEW K-POP GIRL GROUP LANDS NO. 1 & 2 ON WORLD DIGITAL SONGS CHART

Blackpink is less than a month old, but the new girl group is already making major moves on the Billboard charts with their debut singles "Boombayah" and "Whistle."

By **Jeff Benjamin** ▾

Global Rise of K-pop

- Rapid expansion into Western markets (Hot 100, Global 200).
- Increasing visibility of multilingual lyrics (Korean + English).
- English used for “global vibe,” fun indexing, fan connection.
- Music videos + sonic experimentation → global appeal.

Why Study Code-Switching in K-pop?

- English–Korean code switching has become a norm, not an exception.
 - *Frequent code-switch events (≈ 17 – 25 per song depending on groups/solo acts)*
- Lack of **computational** analysis of this phenomenon.
- **Research questions:**
 - *Do male vs. female performers differ in English/Korean usage?*
 - *Does code-switching differ by gender or chart type?*
 - *Can performer gender be predicted purely from lyrics?*

CODE SWITCHING VS CODE MIXING

Definitions

- Code-Switching (CS)
 - *Alternating between languages at boundaries (line, phrase, clause).*
- Code-Mixing
 - *Blending languages within phrases/words.*
- K-pop lyrics typically show:
 - *Inter-sentential switching (“Korean verse → English pre-chorus”)*
 - *Intra-sentential switching (“Hangul + English words together”)*
 - *Tag-switching (“baby,” “yeah,” “okay,” “let’s go”)*

Example

미쳐 that candy's sassy
michyeo

ko-ko

en-en

뉴런의 체계를 갱신

nyuleon-ui

chegyeleul

gaengsin

ko-ko

Why CS Matters for K-pop

- Used for rhythm, emphasis, hooks.
- English often marks “global-ness.”
- Can influence chart performance and cultural indexing.

DATA COLLECTION

Artist & Song Selection

- Time period: Jan 2017 – May 2025.
- Sources:
 - *Lyrics* → *Genius* (via *LyricsGenius*).
 - *Chart data* → *Billboard Hot 100* + *Global 200*.
- 22 artists total:
 - *Groups: 7 female, 4 male*
 - *Soloists: 4 female, 6 male*
- All songs the artists had on either chart (hot 100 or global 200) were retrieved.

Dataset Construction

- Final dataset includes:
 - *Title*
 - *Artist gender (M/F)*
 - *Category (Group/Solo)*
 - *Peak position*
 - *Weeks on chart*
 - *Chart trajectory (time series)*
 - *Raw + cleaned lyrics*
- Separate files for Hot 100 and Global 200.

Dataset Stats

- Most represented:
 - *BTS (29 songs), BLACKPINK (24).*
- Soloists:
 - *V (15), Jungkook (14), Jennie (14).*
- Nearly all artists appear on Global 200;

STATISTICAL TESTS

Feature Engineering for classification

- Korean Lyric Ratio
- English Lyric Ratio
- Code-switch count
- Repetition count
- Alliteration
- Filler words (“oh,” “yeah,” “la,” etc.)
- Combined with multilingual embeddings for the classification task.

Statistical Methods

- Normality check → Shapiro–Wilk
 - This revealed that most features were non-normal.
- Welch's t-test
 - This test showed us unequal variance.
- Proceeded with a Mann–Whitney U for a nonparametric test.
- Set the Significance threshold to $\alpha = 0.05$.

Key Hypotheses

1. Boy vs. Girl Groups → English/Korean ratio differences?
2. Gender differences in number of code-switches?
3. Soloists vs groups?
4. Hot 100 vs Global 200 linguistic differences?

FINDINGS

Findings — Classification

- Boy vs girl groups:
 - *Hot 100*: 76% accuracy and a macro F1-score of 0.76
 - *Global 200*: 72% accuracy and a macro F1-score of 0.71.
- Conclusion
 - Although individual linguistic features (such as the Korean lyric ratio) did not yield significant differences across genders, their combination with learned representations allowed models to predict performer gender at a level significantly above chance.

Findings — Code-Switching Frequency

- Boy vs girl groups:
 - *Hot 100: 19.05 vs 16.96 switches → NOT significant.*
 - *Global 200: 20.57 vs 25.32 → NOT significant.*
- Conclusion: High code-switching across all categories; gender doesn't matter.

Findings — Korean vs English Ratio

- Trend:
 - *Boy groups use **slightly** more Korean.*
 - *Girl groups use **slightly** more English.*
- But Mann–Whitney: NOT statistically significant, therefore we need more data and variety of groups to study on.
- Female soloists, however, are far more English-heavy in their songs (generally >95% in some tracks).

Chart Context Effects

- Boy groups: No difference Hot100 vs G200 → stable Korean ratio.
- Girl groups: Slight trend toward more English on Hot 100 (however not statistically significant).
 - *Hot 100 is a US-centric chart.*
 - *This could suggest a strategic push for English usage for U.S.-oriented releases.*

Interpretation

- English dominance consistent across gender and chart.
- Trends exist but lack statistical power due to less data and limited time period.
- Strong stylistic similarity across groups → English as a global identity marker.

CONCLUSION AND FUTURE DIRECTIONS



Main Takeaways

- English has become a dominant linguistic component of global K-pop.
- Both male and female acts code-switch heavily and frequently.
- Gender differences in language ratio exist but are not statistically significant.
- Classification models achieve up to 76% macro F1, showing latent stylistic cues exist beyond raw language ratios.

Contributions

- First computational analysis linking:
 - *Code-switching*
 - *Billboard chart context*
- Novel dataset combining chart metadata + multilingual lyrics.

Limitations

- Only internationally successful songs → bias.
- Small sample size.
- No audio/visual context (important for K-pop's multimodality).
- Simplifies linguistic structure of Korean.

Future Work

- Expand dataset (domestic charts, pre-2017, smaller agencies).
- Include multimodal features (audio, choreography, MV aesthetics).
- Chart-success prediction using linguistic + non-linguistic features.
- Collaboration with Korean linguists for deeper semantic analysis.
- Temporal analysis: English rise over time.
- Listener reception analysis across markets.

THANK YOU

Send me an email or dog pictures or playlists

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<https://callmesanfornow.github.io>